EXPERMENT -3

CONVOLUTION OF CONTINUOUS TIME AND DISCRETE TIME SIGNALS

TASK A : DISCRETE (Input Coding)

```
clc;
clear;
close all;

%% Discrete-Time Convolution

% Input sequence
x = [1 1 1 1];

% Impulse response
h = [1 1 1 1];

% Convolution
y = conv(x,h);

% Sample indices
nx = 0:length(x)-1;
nh = 0:length(h)-1;
ny = 0:length(y)-1;

%% Plotting
figure;
subplot(3,1,1)
stem(nx,x,'filled','LineWidth',1.5);
grid on;
title('Input Sequence x[n]');
xlabel('n');
ylabel('Amplitude');

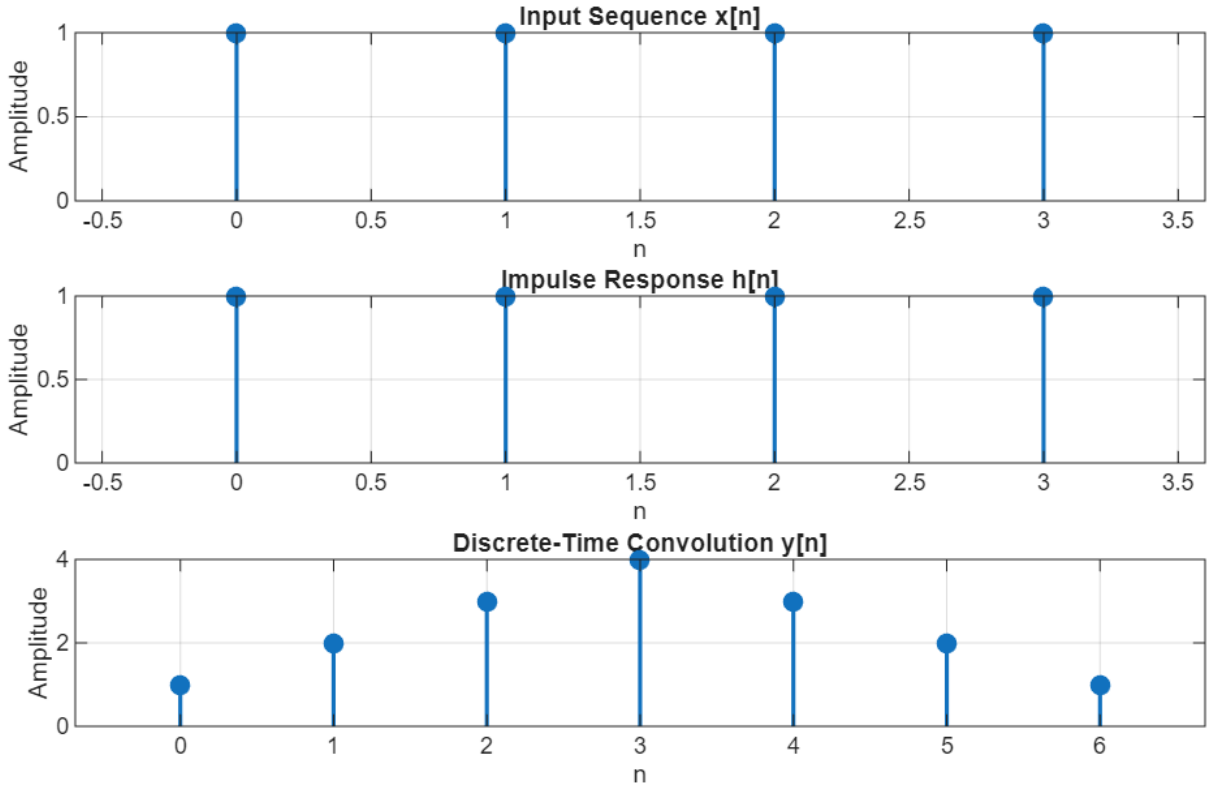
subplot(3,1,2)
stem(nh,h,'filled','LineWidth',1.5);
```

```
grid on;
title('Impulse Response h[n]');
xlabel('n');
ylabel('Amplitude');

subplot(3,1,3)
stem(ny,y,'filled','LineWidth',1.5);
grid on;
title('Discrete-Time Convolution y[n]');
xlabel('n');
ylabel('Amplitude');

disp('Convolution Output:');
disp(y);
```

Output




```
Convolution Output:
     1     2     3     4     3     2     1
```

CONTINUOUS (Input coding)

```
clc;
clear;
close all;

%% Continuous-Time Convolution

dt = 0.01;               % Sampling interval
t = 0:dt:1;              % Time axis (Pulse duration = 1 sec)

% Input signal x(t)
x = ones(size(t));

% Impulse response h(t)
h = ones(size(t));

% Convolution
y = conv(x,h)*dt;

% Time axis for output
ty = 0:dt:2;

%% Plotting

figure;

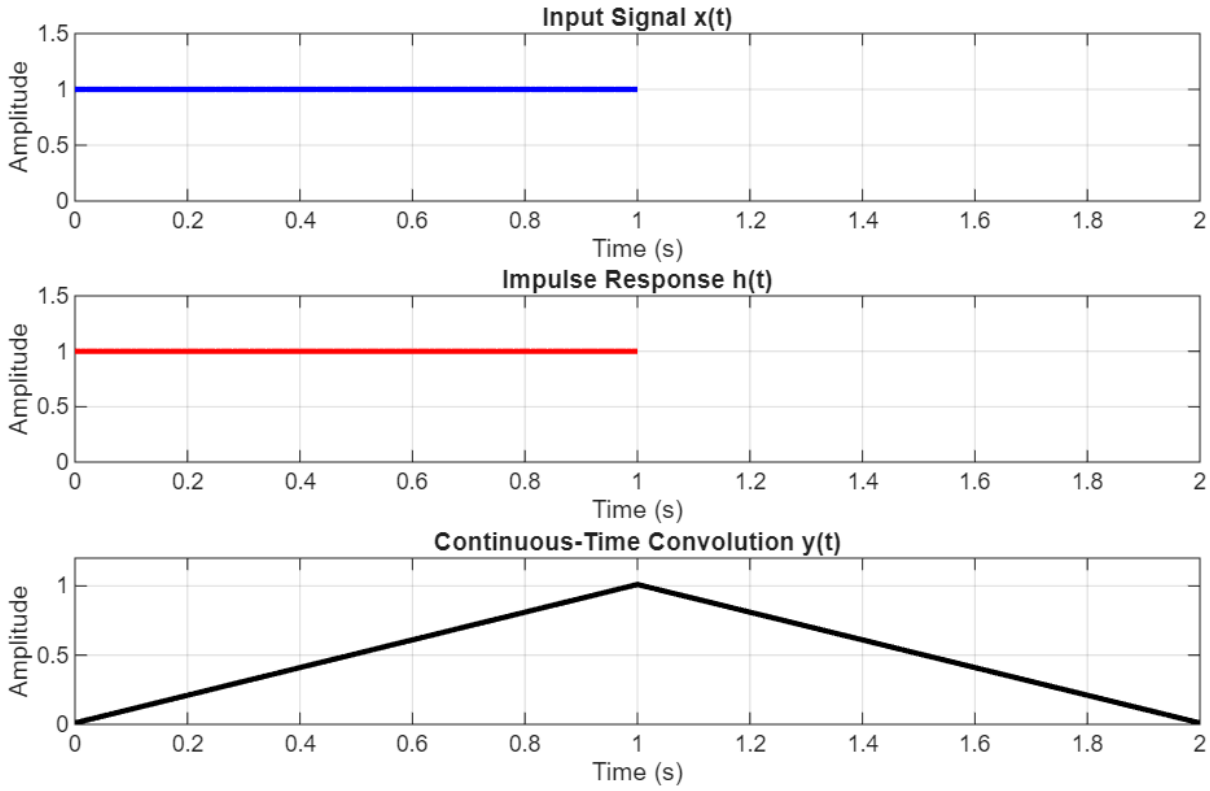
subplot(3,1,1)
plot(t,x,'b','LineWidth',2);
grid on;
title('Input Signal x(t)');
xlabel('Time (s)');
ylabel('Amplitude');
axis([0 2 0 1.5]);

subplot(3,1,2)
plot(t,h,'r','LineWidth',2);
grid on;
title('Impulse Response h(t)');
xlabel('Time (s)');
ylabel('Amplitude');
axis([0 2 0 1.5]);

subplot(3,1,3)
plot(ty,y,'k','LineWidth',2);
grid on;
title('Continuous-Time Convolution y(t)');
xlabel('Time (s)');
ylabel('Amplitude');
axis([0 2 0 1.2]);

disp('Output Duration = 2 seconds');
disp('Peak occurs at t = 1 second');
```

OUTPUT




```
Output Duration = 2 seconds
Peak occurs at t = 1 second
```